# The Satake Isomorphism made explicit

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## Contents



These are my notes for my talks in the Satake isomorphism seminar about explicitly determining the Satake transform. My thanks to Lucas Mann for his guidance while preparing these talks. Comments/corrections are welcome at shyam.ravishankar@uni-muenster.de.

Our goal is to explicitly describe the Satake isomorphism

$$\mathcal{S}\colon \mathcal{H}(G,K)\otimes_{\mathbf{Z}}\mathbf{Z}[q^{1/2},q^{-1/2}]\xrightarrow{\sim} R(\hat{G})\otimes_{\mathbf{Z}}\mathbf{Z}[q^{1/2},q^{-1/2}].$$

Fixed notation for the seminar following [Gro98]: Let $F$ be a local field, $\underline{G}$ be a reductive group over $\mathcal{O}_F$ which is split over $F$, $\underline{T}\subset\underline{B}\subset\underline{G}$ a split maximal torus and Borel, $\underline{N}\subset\underline{B}$ its unipotent radical. $X^*(\underline{T}), X_*(\underline{T})$ are the characters and cocharacters, $W=N_G(\underline{T})/\underline{T}$ the Weyl group, $\Phi,\Phi^+,\Delta$ the roots, positive roots, and simple roots. The symbol $\cdot^\vee$ represents the dual data to $\cdot$ (e.g. $\Phi^\vee$ the coroots). Hats denote the dual groups over $\mathbf{C}$ i.e. $\hat{G}$ is the complex reductive group dual to $\underline{G}$ and $\hat{T}\subset\hat{B}\subset\hat{G}$ are the associated split maximal torus and Borel to the fixed $\underline{T}\subset\underline{B}\subset\underline{G}$.

We also have the locally compact groups $G=\underline{G}(F), K=\underline{G}(\mathcal{O}_F), T=\underline{T}(F), B=\underline{B}(F), N=\underline{N}(F)$. Fix $\pi\in\mathcal{O}_F$ a uniformizer.

**Example:**

Examples will be in a box like this. The canonical example of a reductive group is $\underline{G}=\mathrm{GL}_n$. Later this will not be a valid/useful example so some other examples are $\mathrm{SL}_n$, $\mathrm{PGL}_n$, and $\mathrm{PSp}_{2n}$. For $\underline{G}=\mathrm{GL}_n$ the standard choices of split maximal torus and Borel are the diagonal and upper triangular matrices, respectively. These give in the obvious way choices of torus and Borel in $\mathrm{SL}_n$ and $\mathrm{PGL}_n$.
With this choice of torus and Borel for $\mathrm{GL}_n$, the unipotent radical $\underline{N}$ of $\underline{B}$ is the subgroup of unipotent upper triangular matrices. The normalizer $N_G(\underline{T})$ is the subgroup of monomial matrices (matrices with exactly one nonzero entry in each row and each column), and the Weyl group $W=N_G(\underline{T})/\underline{T}$ can be identified with the group of permutation matrices, hence with $S_n$. The actions of $W=S_n$ on $X^*(\underline{T})$ and $X_*(\underline{T})$ are given simply by permuting coordinates.

We fix some notation/terminology. Note that the coweights of $\underline{G}$ are the same as the weights of $\hat{G}$. More precisely, we can identify $X_*(\underline{T})\simeq X^*(\hat{T})$. Hence we can identify pretty much any dual notion for $\underline{G}$ as the normal notion for $\hat{G}$

(e.g. the dominant coweights $P^+$ are the dominant weights for $\hat{G}$). To minimize confusion we will fix the following convention for the rest of the talk: there will be no 'co.' E.g. we refer to the cocharacters $X_*(\underline{T})$ of $\underline{T}$ as the characters $X^*(\hat{T})$ of $\hat{T}$ and refer to the coroots of $\underline{G}$ as the roots of $\hat{G}$.

# 1 Recollections on reductive groups and their representation theory

Recall we had the Cartan decomposition:

$$G = \bigsqcup_{\lambda \in P^+} K\lambda(\pi)K$$

where $P^+$ is the set of **dominant coweights** (which from now on we will call the **dominant weights** of $\hat{G}$)

$$P^+ = \{\lambda \in X_*(\underline{T})\colon \langle \lambda, \alpha \rangle \geq 0 \text{ for all } \alpha \in \Phi^+\} = \{\lambda \in X_*(\underline{T})\colon \langle \lambda, \alpha \rangle \geq 0 \text{ for all } \alpha \in \Delta\}.$$

**Example:** $\mathrm{GL}_n$

Writing $X_*(\underline{T}) = \langle f_1, \dots, f_n \rangle$, we have

$$P^+ = \left\{ \sum r_i f_i : r_1 \geq r_2 \geq \dots \geq r_n \right\}.$$

This implies that the Hecke algebra $\mathcal{H}(G, K)$ is generated as a $\mathbf{Z}$-module by the characteristic functions

$$c_\lambda = \mathrm{char}(K\lambda(\pi)K)$$

for $\lambda \in P^+$. Hence to explicitly determine the Satake isomorphism boils down to explicitly determining these $\mathcal{S}c_\lambda$. Let's try to simplify a bit further.

A **set of fundamental weights**[1] for $\hat{G}$ is a subset $FW \subset P^+$ such that

1. $FW$ are $\mathbf{Z}$-linearly independent in $X^*(\hat{T})$;
2. For each $\lambda \in FW$, $\langle \lambda, \alpha \rangle$ is nonzero for exactly one $\alpha \in \Delta$;
3. If $\langle \lambda, \alpha \rangle \neq 0$ for some $\lambda \in FW$, $\alpha \in \Delta$, then $\langle \lambda, \alpha \rangle = 1$.

One should think of fundamental weights as a sort of dual basis for $X^*(\hat{T})$ which is dual to the basis $\Delta$ of $X^*(\underline{T})$. We note that $\Delta$ is not necessarily an actual basis of $X^*(\underline{T})$ and hence a set of fundamental weights is not necessarily unique since we can always scale those $\lambda$ which satisfy $\langle \lambda, \alpha \rangle = 1$ for some $\alpha \in \Delta$ by an element of

$$X_0 := \left\{ \mu \in X^*(\hat{T})\colon \langle \mu, \alpha \rangle = 0 \, \forall \alpha \in \Delta \right\}.$$

Given a fixed set of fundamental weights, it should be clear that we can write any $P^+$ uniquely as a linear combination of fundamental weights, with nonnegative coefficients for fundamental weights not in $X_0$. This means that to understand the Satake transform $\mathcal{S}$, it is in principle[2] enough to understand it on fundamental weights as well as $X_0$. We fix now a set of fundamental weights $FW$. From now on we will abuse terminology and refer to this choice of set of fundamental weights as "the fundamental weights of $\hat{G}$."

A **miniscule weight** for $\hat{G}$ is a dominant weight $\lambda \in P^+$ which is minimal in $P^+$ for the partial ordering on $X^*(\hat{T}) = X_*(\underline{T})$ given by

$$\mu \leq \nu \qquad \Longleftrightarrow \qquad \nu - \mu = \sum_{i=1}^{k} \alpha_i$$

[1] The experienced reader may realize that this is not really a correct definition of fundamental weight, but it suffices for our purposes.

[2] The situation is a bit more complicated than this since $\mathcal{H}(G, K)$ is not a free algebra on the fundamental weights; however we still have

$$c_\lambda c_\mu = c_{\lambda+\mu} + \sum_{\nu < \lambda+\mu} n_{\lambda,\mu}(\nu) c_\nu,$$

(eqn. 2.9 in [Gro98]), so one can still envision first determining $\mathcal{S}c_\lambda$ for the miniscule weights and then systematically computing up through the (order-theoretic) lattice $P^+$.

for some (not necessarily distinct) simple coroots $\alpha_i \in \Delta^\vee$.

**Example:** $\mathrm{GL}_n$

A set of fundamental weights for $\hat{G} = \mathrm{GL}_n$ are

$$\left\{ \sum_{i=1}^{j} f_i \colon 1 \le j < n \right\}.$$

We take this as our fixed set $FW$. The "kernel" of the pairing is

$$X_0 = \left\{ \sum_{i=1}^{n} r f_i \colon r \in \mathbf{Z} \right\}.$$

One can try to check now that these are miniscule weights for $\mathrm{GL}_n$. Later we will see a nice representation-theoretic criterion which easily demonstrates this fact.

## 1.1 Diversion into Representation Theory

The genesis of all these data and notions is to understand representation theory of $\underline{G}$. We will describe here the relevant translation of all this into representation-theoretic terms, as we will need this for understanding the Satake isomorphism.

First we start with the following question: Why is $\mathbf{Z}[X^*(\hat{T})]^W$ called $R(\hat{G})$? Let $K_0 \operatorname{Rep}(\hat{G})$ denote the **representation ring** of $\hat{G}$. This is the Grothendieck ring of the category of finite-dimensional representations of $\hat{G}$; more concretely it is comprised of isomorphism classes of $\hat{G}$-representations, with addition given by direct sum and multiplication given by tensor product (and we add in formal inverses for the addition). We shall soon see that $K_0 \operatorname{Rep}(\hat{G})$ is isomorphic to $\mathbf{Z}[X^*(\hat{T})]^W$, which justifies our notation $R(\hat{G})$.

Any representation $V$ of $\hat{G}$ decomposes as a direct sum of 1-dimensional $\hat{T}$-representations. We can group these to obtain an eigenspace decomposition

$$V = \bigoplus_{\mu} V_\mu$$

of $\hat{T}$-representations, where each $V_\mu$ is a subspace on which $\hat{T}$ acts by the character $\mu \in X^*(\hat{T})$. We call these $V_\mu$ the **weight spaces of $V$** and the associated $\mu$ the **weights of $V$**. A weight $\mu$ of $V$ is called a **highest weight** if all other weights $\lambda$ of $V$ satisfy $\lambda \le \mu$ in the partial order defined on $X^*(\hat{T})$.

**Example:** The adjoint representation and the (positive) roots

By definition, the roots of $\hat{G}$ are (along with 0) the weights of the adjoint representation of $\hat{G}$. More precisely, the adjoint representation of $\hat{G}$ on its Lie algebra $\hat{\mathfrak{g}}$ decomposes into

$$\hat{\mathfrak{g}} = \hat{\mathfrak{t}} \oplus \bigoplus_{\alpha \in \Phi^\vee} L_\alpha$$

where $\hat{\mathfrak{t}}$, the Lie algebra of the torus $\hat{T}$, is the weight space with weight 0 and the $L_\alpha$ (which are all nontrivial for $\alpha \in \Phi^\vee$!) are weight spaces with nontrivial weight.
Applying this same construction to the chosen Borel $\hat{B} \subset \hat{G}$ yields the positive roots:

$$\hat{\mathfrak{b}} = \hat{\mathfrak{t}} \oplus \bigoplus_{\alpha \in \Phi^{\vee +}} L_\alpha.$$

We define the **character** of $V$ by

$$\chi_V \coloneqq \sum_{\mu \in X^*(\hat{T})} m_V(\mu)[\mu] \in \mathbf{Z}[X^*(\hat{T})]$$

where $m_V(\mu)$ is the dimension of the weight space $V_\mu$ in the decomposition

$$V = \bigoplus_\mu V_\mu.$$

This defines a ring homomorphism $\chi\colon K_0 \operatorname{Rep}(\hat{G}) \to \mathbf{Z}[X^*(\hat{T})]$.

**Proposition 1**

The homomorphism $\chi$ is injective with image equal to the Weyl invariants $\mathbf{Z}[X^*(\hat{T})]^W$.

So this gives us another description of the ring $R(\hat{G})$, and we can go further by describing a $\mathbf{Z}$-basis for this ring.

**Theorem 1: Theorem of the Highest Weight**

Let $V$ be an irreducible $\hat{G}$-representation.

1. There is a unique one-dimensional $\hat{B}$-stable subspace $L \subset V$, which is a weight space for $\hat{T}$.
2. Call the $\hat{B}$-stable weight space $L = V_\mu$ for a character $\mu \in X^*(\hat{T})$. Then $\mu \in P^+$.
3. $\mu$ is the unique highest weight of $V$.
4. The map $V \mapsto \mu$ from the set of isomorphism classes of irreducible $\hat{G}$-representations to $P^+$ is a bijection. That is, any two irreps with the same highest weight are isomorphic, and given any $\mu \in P^+$ there is an irrep with highest weight $\mu$.

In characteristic $0$, all $\hat{G}$-representations are direct sums of irreps; hence we have an isomorphism of $\mathbf{Z}$-modules $R(\hat{G}) \simeq \bigoplus_{P^+} \mathbf{Z}$. From this alone we see that $R(\hat{G}) \simeq \mathcal{H}(G, K)$ as $\mathbf{Z}$-modules!

We can in fact say more. We call an irreducible representation **fundamental** if its highest weight is a fundamental weight. This is again as unique as the set of fundamental weights, and we will specifically deign to only call those representations fundamental which have highest weight in our fixed set of fundamental weights.

**Example:** $\mathrm{GL}_n$

The fundamental representations of $\mathrm{GL}_n$ (for our fixed choice of fundamental weights) are the exterior powers $\bigwedge^i \mathbf{C}^n$ for $1 \le i < n$. Here $\mathbf{C}^n$ is the canonical representation $\mathrm{id}\colon \mathrm{GL}_n \to \mathrm{GL}_n$. These decompose as such:

$$\bigwedge^i \mathbf{C}^n = \bigoplus_{k_1 < \cdots < k_i} \mathbf{C}(e_{k_1} \wedge \cdots \wedge e_{k_i})$$

where $\{e_1, \dots, e_n\}$ is the standard basis of $\mathbf{C}^n$. The weight of $\mathbf{C}(e_{k_1} \wedge \cdots \wedge e_{k_i})$ is

$$f_{k_1} + \cdots + f_{k_i},$$

and hence the highest weight is

$$f_1 + \cdots + f_i,$$

which is fundamental.

**Remark: Generating irreps from fundamental reps**

The fact that the fundamental weights (with $X_0$) generate $P^+$ is reflected representation-theoretically by a construction of Cartan which realizes arbitrary irreducible representations as subrepresentations of appropriate tensor products of fundamental representations. The idea is that if $\mu \in P^+$ can be written as

$$\mu = \sum_{\lambda \in FW} k_\lambda \lambda$$

then we should search for the corresponding irreducible representation $V_\mu$ inside the representation

$$W = \bigotimes_{\lambda \in FW} V_\lambda^{\otimes k_\lambda}.$$

It is easy enough to see that since tensor products commute with direct sums, the weight space decomposition of $W$ comes from taking tensor products of the weight spaces of the individual $V_\lambda$, and hence the weights of $W$ are sums of the weights of the $V_\lambda$. In particular, $W$ will have highest weight $\mu$; the only issue is that $W$ will in general not be irreducible. To fix this problem, one can find a suitable quotient or (equivalently[a]) subrepresentation which is irreducible and has the same weights.

We demonstrate this with an example for $\mathrm{GL}_n$: in this case, $X_0$ is generated by $f_1 + \cdots + f_n$, which corresponds to the irreducible representation $\bigwedge^n \mathbf{C}^n$. We observe that this representation is given by $\det\colon \mathrm{GL}_n \to \mathrm{GL}_1$ and that correspondingly the character $f_1 + \cdots + f_n$ is precisely $\det$ restricted to $\hat{T}$. As the determinant representation is 1-dimensional, it makes sense to define negative tensor powers

$$\left(\bigwedge^n \mathbf{C}^n\right)^{\otimes -k} := \left(\left(\bigwedge^n \mathbf{C}^n\right)^*\right)^{\otimes k}$$

where

$$\left(\bigwedge^n \mathbf{C}^n\right)^* := \mathrm{Hom}_{\mathbf{C}}\left(\bigwedge^n \mathbf{C}^n, \mathbf{C}\right)$$

is the dual. Then scaling an element of $P^+$ by $k(f_1+\cdots+f_n)$ precisely corresponds to tensoring the corresponding irreducible representation by $(\bigwedge^n \mathbf{C}^n)^{\otimes k}$.

Let's work now with $\mathrm{GL}_2$; we want to construct the irreducible representation with highest weight $2f_1 \in P^+$. In our particular example, $f_1$ is fundamental so we should examine the representation $\mathbf{C}^2 \otimes \mathbf{C}^2$. Let $\{x_1, x_2\}$ be the standard basis for the first copy of $\mathbf{C}^2$, and let $\{y_1, y_2\}$ be the standard basis for the second copy. Then as $\hat{T}$-representations we have a decomposition

$$\mathbf{C}^2 \otimes \mathbf{C}^2 = \mathbf{C}(x_1 \otimes y_1) \oplus \mathbf{C}(x_2 \otimes y_2) \oplus \bigoplus_{i \neq j} \mathbf{C}(x_i \otimes y_j).$$

These are weight spaces with weights $2f_1$, $2f_2$, and $f_1 + f_2$, respectively; hence $\mathbf{C}^2 \otimes \mathbf{C}^2$ has unique highest weight $2f_1$, as we wanted. However, this is not an irreducible $\hat{G}$-representation! We can find in the last factor the subspace

$$\mathbf{C}(x_1 \otimes y_2 - x_2 \otimes y_1),$$

which is $\hat{G}$-stable and on which $\hat{G}$ acts by $\det$. The quotient of $\mathbf{C}^2 \otimes \mathbf{C}^2$ by this subrepresentation is by definition the symmetric power $\mathrm{Sym}^2 \mathbf{C}^2$, and by choosing a complement of this subspace in

$$\bigoplus_{i \neq j} \mathbf{C}(x_i \otimes y_j)$$

we realize $\mathrm{Sym}^2 \mathbf{C}^2$ as a subrepresentation of $\mathbf{C}^2 \otimes \mathbf{C}^2$. This is irreducible, and its weights are $2f_1$, $2f_2$, and $f_1 + f_2$; but now the weight space of $f_1 + f_2$ has dimension 1. Hence $\mathrm{Sym}^2 \mathbf{C}^2$ is the irreducible representation of $\mathrm{GL}_2$ with highest weight $2f_1$ that we were searching for.

[a] Since we are in characteristic 0, the category $\mathrm{Rep}(\hat{G})$ is semisimple, so every exact sequence splits!

We also quickly finish by giving an interpretation of the miniscule weights:

**Lemma 1**

Let $\lambda \in P^+$ and let $V_\lambda$ be the unique irrep with highest weight $\lambda$. Then $\lambda$ is miniscule if and only if $W$ acts transitively on the weights of $V_\lambda$.

**Example:** $\mathrm{GL}_n$

It is very easy to see from this criterion that the fundamental weights of $\mathrm{GL}_n$ are all miniscule: the weights of $\bigwedge^i \mathbf{C}^n$ are

$$\left\{f_{k_1} + \cdots + f_{k_i} : k_1 < \cdots < k_i\right\},$$

and clearly these are within the same orbit of the permutation action of $W = S_n$.

# 2 Computing the Satake Transform

Recall the isomorphism $\mathcal{H}(T, K \cap T) \xrightarrow{\sim} \mathbf{Z}[X^*(\hat{T})]$ given by

$$\mathrm{char}(\nu(\pi)(K \cap T)) \mapsto [\nu];$$

under this isomorphism we can write $\mathcal{S}c_\lambda$ in the canonical $\mathbf{Z}[q^{1/2}, q^{-1/2}]$-basis of $\mathbf{Z}[X^*(\hat{T})] \otimes \mathbf{Z}[q^{1/2}, q^{-1/2}]$

$$\mathcal{S}c_\lambda = \sum_{\nu \in X^*(\hat{T})} \mathcal{S}c_\lambda(\nu(\pi))[\nu].$$

We saw in the proof that the Satake transform is an isomorphism that those $\mu \in P^+$ which satisfy $\mathcal{S}c_\lambda(\mu(\pi)) \neq 0$ also satisfy $\mu \leq \lambda$, and that $\mathcal{S}c_\lambda(\lambda(\pi)) \neq 0$. One can even show that

$$\mathcal{S}c_\lambda(\lambda(\pi)) = q^{\langle \lambda, \rho \rangle},$$

where

$$\rho := \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha.$$

Let's rewrite these coefficients in a different way now.

For each $\mu \in P^+$, let $V_\mu$ be the unique (up to isomorphism) irreducible $\hat{G}$-representation with highest weight $\mu$, and let $\chi_\mu \in \mathbf{Z}[X^*(\hat{T})]^W = R(\hat{G})$ denote the character of the representation $V_\mu$. Then the $\chi_\mu$ form a $\mathbf{Z}$-basis of $R(\hat{G})$, so for each $\lambda \in P^+$ we can uniquely write

$$\mathcal{S}c_\lambda = \sum_{\mu \in P^+} a_\lambda(\mu) \chi_\mu$$

in $R(\hat{G}) \otimes \mathbf{Z}[q^{1/2}, q^{-1/2}]$. To compute the Satake transform thus also amounts to computing the coefficients $a_\lambda(\mu)$. Remarkably, what we know about the coefficients $\mathcal{S}c_\lambda(\mu(\pi))$ also applies to the $a_\lambda(\mu)$.

**Lemma 2**

If $a_\lambda(\nu) \neq 0$ for $\nu \in P^+$ then $\nu \leq \lambda$. Moreover,

$$a_\lambda(\lambda) = \mathcal{S}c_\lambda(\lambda(\pi)) = q^{\langle \lambda, \rho \rangle};$$

hence we can write

$$\mathcal{S}c_\lambda = q^{\langle \lambda, \rho \rangle} + \sum_{\mu < \lambda} a_\lambda(\mu) \chi_\mu.$$

*Proof.* Write the character $\chi_\mu$ in $\mathbf{Z}[X^*(\hat{T})]$ as

$$\chi_\mu = \sum_{\nu \in X^*(\hat{T})} m_\mu(\nu).$$

With this we can rewrite our first expression for the Satake transform as

$$\mathcal{S}c_\lambda = \sum_{\nu \in X^*(\hat{T})} \left( \sum_{\mu \in P^+} a_\lambda(\mu) m_\mu(\nu) \right) [\nu],$$

which yields the formula

$$\mathcal{S}c_\lambda(\nu(\pi)) = \sum_{\mu \in P^+} a_\lambda(\mu) m_\mu(\nu). \tag{1}$$

Assume $a_\lambda(\nu) \neq 0$ for some $\nu \in P^+$ which does not satisfy $\nu \leq \lambda$. We consider equation (1) and note that **theorem 1** implies the following:

- $m_\mu(\nu) \neq 0 \implies \mu \geq \nu$;
- $m_\nu(\nu) = 1$.

By assumption on $\nu$ we know $\mathcal{S}c_\lambda(\nu(\pi)) = 0$, hence

$$a_\lambda(\nu) = -\sum_{\mu > \nu} a_\lambda(\mu) m_\mu(\nu) \neq 0.$$

In particular we must have some $\nu_1 > \nu$ in $P^+$ such that $a_\lambda(\nu_1) \neq 0$. In this manner we can construct an increasing sequence in $P^+$

$$\nu = \nu_0 < \nu_1 < \nu_2 < \cdots$$

such that $a_\lambda(\nu_i) \neq 0$ for all $i$; but only finitely many $\nu \in P^+$ can satisfy $a_\lambda(\nu) \neq 0$! Hence all $\nu \in P^+$ which satisfy $a_\lambda(\nu) \neq 0$ must also satisfy $\nu \leq \lambda$.

To prove the second part of the claim, we now write

$$\mathcal{S}c_\lambda(\lambda(\pi)) = a_\lambda(\lambda) + \sum_{\mu > \lambda} a_\lambda(\mu) m_\mu(\lambda) = a_\lambda(\lambda)$$

since $a_\lambda(\mu)$ must vanish for $\mu > \lambda$. ■

**Example: $a_\lambda(\mu) \neq \mathcal{S}c_\lambda(\mu)$**

Despite what this lemma might make us want to believe, we in fact do not have $a_\lambda(\mu) = \mathcal{S}c_\lambda(\mu)$ in general for $\lambda, \mu \in P^+$. Specifically for $\widehat{G} = \mathrm{GL}_2$, consider the irreducible representation $\mathrm{Sym}^2 \mathbf{C}^2$; this has character

$$\chi_V = [2e_1] + [2e_2] + [e_1 + e_2].$$

Let $\lambda = 2e_1$ and $\mu = e_1 + e_2$. We can show essentially by brute force that the only $\nu \in P^+$ which satisfies $\nu > \mu$ and $\nu \leq \lambda$ is $\nu = \lambda = 2e_1$. Hence we have

$$\mathcal{S}c_\lambda(\mu(\pi)) = a_\lambda(\mu) + \sum_{\nu > \mu} a_\lambda(\nu) m_\nu(\mu) = a_\lambda(\mu) + a_\lambda(\lambda).$$

In particular we must have $a_\lambda(\mu) \neq \mathcal{S}c_\lambda(\mu(\pi))$ since $a_\lambda(\lambda) \neq 0$.

Let us take now a rescaled basis of $R(\widehat{G}) \otimes \mathbf{Z}[q^{1/2}, q^{-1/2}]$; precisely we take

$$\varphi_\lambda = q^{\langle \lambda, \rho \rangle} \chi_\lambda.$$

These evidently form a $\mathbf{Z}$-basis of the ring, so we have

$$\mathcal{S}c_\lambda = \varphi_\lambda + \sum_{\mu < \lambda} b_\lambda(\mu) \varphi_\mu$$

for *integral* coefficients $b_\lambda(\mu)$. Recalling that $\mathcal{S}c_\lambda$ form a $\mathbf{Z}$-basis for the ring $R(\widehat{G}) \otimes \mathbf{Z}[q^{1/2}, q^{-1/2}]$, we may also write

$$\varphi_\lambda = q^{\langle \lambda, \rho \rangle} \chi_\lambda = \mathcal{S}c_\lambda + \sum_{\mu < \lambda} d_\lambda(\mu) \mathcal{S}c_\mu. \tag{2}$$

Here we again have vanishing of $d_\lambda(\mu)$ for $\mu \not\leq \lambda$ by a generalization of the fact that upper triangular matrices have upper triangular inverse. So to compute the Satake transform, we have reduced to computing either the integers $b_\lambda(\mu)$ or the integers $d_\lambda(\mu)$, for $\lambda, \mu \in P^+$. As mentioned previously, we may even in principle restrict our focus to $\lambda$ which are fundamental weights. We will later see a way to compute the $d_\lambda(\mu)$ as certain polynomials in the case that $\underline{G}$ is "semisimple of adjoint type."

For now, we conclude with an example. If $\lambda \in P^+$ is miniscule (i.e. minimal), then we automatically have $\mathcal{S}c_\lambda = \varphi_\lambda$. As we've seen this happens for $\mathrm{GL}_n$:

**Example:** $\mathrm{GL}_n$

The fundamental weights $f_1 + \cdots + f_i$ for $0 \leq i \leq n$ are all miniscule. Hence we have the following formula, first due to Tamagawa:

$$\mathcal{S}\,\mathrm{char}(K \,\mathrm{diag}(\underbrace{\pi, \ldots, \pi}_{i \text{ times}}, \overbrace{1, \ldots, 1}^{n-i \text{ times}})K) = q^{\frac{i(n-i)}{2}} \chi\left(\bigwedge^i \mathbf{C}^n\right).$$

In particular, for $i = n$ we have

$$\mathcal{S}\,\mathrm{char}(K\pi I K) = \chi\left(\bigwedge^n \mathbf{C}^n\right) = [\det] = [f_1 + \cdots + f_n].$$

# 3 Computing $d_\lambda(\mu)$

The case of $\mathrm{GL}_n$ is in a sense settled, but what happens when we have (fundamental) weights which are not miniscule? In some nice cases, work of Lusztig and S. Kato gives us an answer.

## 3.1 Semisimple groups (of adjoint type)

For Kato's results to apply, we need to assume that $\underline{G}$ is a so-called "semisimple group of adjoint type." For us, a **semisimple group** is a reductive group such that the roots $\Phi$ generate a finite index (equivalently full rank) sublattice of $X^*(\underline{T})$. Over perfect fields, this is equivalent to the center $Z(\underline{G})$ having trivial connected identity component.

A semisimple group is of **adjoint type** if $\Phi$ generates the entire lattice $X^*(\underline{T})$. Equivalently, it is a semisimple group with trivial center. Given a semisimple group $\underline{G}$, the (sheaf-theoretic!) quotient $\underline{G}/Z(\underline{G})$ is a group of adjoint type. Additionally, a semisimple group $\underline{G}$ is of adjoint type if and only if $\widehat{G}$ is **simply connected**; in our case since $\widehat{G}$ is defined over $\mathbf{C}$ this means that $\widehat{G}(\mathbf{C})$ with the analytic topology is simply connected[3].

**Example:**

$\mathrm{GL}_n$ is not semisimple, as it has only $n-1$ simple roots, but the character lattice $X^*(\underline{T})$ is of rank $n$. The group $\mathrm{SL}_n$ is semisimple, however, as it has the same roots as $\mathrm{GL}_n$ and we can view its characters as a quotient of the character lattice $X^*(\underline{T}) = \langle f_1, \ldots, f_n \rangle$ for $\mathrm{GL}_n$

$$X^*(\underline{T}) / \langle f_1 + \cdots + f_n \rangle.$$

[3]There is also a more general definition of simply connected semisimple group which works over any field; it is still true in this generality that a semisimple group is simply connected if and only if its dual is of adjoint type.

Moreover $\mathrm{SL}_n$ is simply connected, as its dual $\mathrm{PGL}_n$ has character lattice

$$\left\{ r_1 e_1 + \cdots + r_n e_n \in X^*(\underline{T}) \colon \sum_{i=1}^{n} r_i = 0 \right\} \subset X^*(\underline{T}),$$

which is generated by the roots $\Phi = \{e_i - e_{i+1}\}$.
The projective general linear group $\mathrm{PGL}_n$ is defined as $\mathrm{GL}_n / Z(\mathrm{GL}_n)$. Be warned that this is a quotient of fppf sheaves, so we do not in general have $\mathrm{PGL}_n(R) = \mathrm{GL}_n(R)/\mathbf{G}_m(R)$ (though this is true when $R$ is a field). Moreover, if we define $\mathrm{PSL}_n := \mathrm{SL}_n / Z(\mathrm{SL}_n)$, it turns out that the natural morphism $\mathrm{PSL}_n \hookrightarrow \mathrm{PGL}_n$ is an isomorphism!

Assume from now on that $\underline{G}$ is semisimple of adjoint type.

## 3.2 Statement of the main result

For any $\mu \in X^*(\hat{T})$, let

$$\hat{P}(\mu) := \sum_{\substack{\mu = \sum n(\alpha^\vee)\alpha^\vee \\ n(\alpha^\vee) \in \mathbf{Z}_{\geq 0} \\ \alpha \in \Phi^+}} q^{-\sum n(\alpha^\vee)}$$

be the generating function in $q^{-1}$ for expressions of $\mu$ as a non-negative sum of positive roots. If $\mu$ is not expressible as such a sum, then by convention we have $\hat{P}(\mu) = 0$. We also have $\hat{P}(0) = 1$. One can also show that $\langle \alpha^\vee, \rho \rangle = 1$ for $\alpha \in \Delta$; hence for $\mu \geq 0$ we have $\langle \mu, \rho \rangle \in \mathbf{Z}_{\geq 0}$ and moreover

$$q^{\langle \mu, \rho \rangle} \hat{P}(\mu)$$

is a polynomial in $q$ with $\mathbf{Z}$-coefficients. As $\hat{P}(\mu) = 0$ otherwise the same statement holds for all $\mu \in X^*(\hat{T})$. Set

$$\rho^\vee := \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha^\vee.$$

**Proposition 2: (Kato)**

The coefficients $d_\lambda(\mu)$ are given by

$$d_\lambda(\mu) = P_{\mu,\lambda}(q) = q^{\langle \lambda - \mu, \rho \rangle} \sum_{\sigma \in W} \varepsilon(\sigma) \hat{P}(\sigma(\lambda + \rho^\vee) - (\mu + \rho^\vee)),$$

where $\varepsilon(\sigma) := \det(\sigma | X^*(\hat{T}))$ is the sign character on $W$.

**Remark: Kazhdan–Lusztig polynomials**

The polynomial $P_{\mu,\lambda}$ in the proposition is a Kazhdan–Lusztig polynomial for the affine Weyl group of $\hat{G}$. See Kato's paper [Kat82] for more; note that Kato is working with an abstract root system rather than a reductive group. To get Kato's results in our setting, take the root system of $\hat{G}$ – the assumption that $\underline{G}$ is semisimple of adjoint type translates to the weight lattice $P$ in Kato's setting being exactly $X^*(\hat{T})$.

**Example:** $\mathrm{PGL}_2$

For now we give a couple of short examples in the case of $\underline{G} = \mathrm{PGL}_2$; in this case we have

- $\hat{G} = \mathrm{SL}_2$,
- $W = S_2 = \{\mathrm{id}, \sigma\}$,

- $\rho^\vee = \frac{1}{2}(f_1 - f_2)$.

First we consider $\lambda = f_1$, $\mu = f_1 + f_2 = 0$; since $\lambda$ is miniscule we should expect that **proposition 2** gives $d_\lambda(\mu) = 0$. Indeed, we can compute

$$\mathrm{id}(\lambda + \rho^\vee) - (\mu + \rho^\vee) = \lambda - \mu = \lambda$$

and $\lambda \not\geq 0$ i.e. $\hat{P}(\lambda) = 0$. Moreover,

$$\sigma(\lambda + \rho^\vee) - (\mu + \rho^\vee) = f_2 + \frac{1}{2}(f_2 - f_1) - \frac{1}{2}(f_1 - f_2) = 2f_2 - f_1 = 3f_2$$

and again $\hat{P}(3f_2) = 0$; hence $d_\lambda(\mu) = 0$ as expected.

We also compute $d_\lambda(\mu)$ for $\lambda = 2f_1$, $\mu = f_1 + f_2 = 0$; in this case $\lambda - \mu = f_1 - f_2$ i.e. $\lambda \geq \mu$, so we don't a priori know what $d_\lambda(\mu)$ is supposed to be. First we find that

$$\mathrm{id}(\lambda + \rho^\vee) - (\mu + \rho^\vee) = \lambda - \mu = f_1 - f_2 \geq 0,$$

and since there is only one simple root $(f_1 - f_2)$, the generating function $\hat{P}$ behaves as a partition function. Hence $\hat{P}(\lambda - \mu) = q^{-1}$. On the other hand,

$$\sigma(\lambda + \rho^\vee) - (\mu + \rho^\vee) = 2f_2 + \frac{1}{2}(f_2 - f_1) - \frac{1}{2}(f_1 - f_2) = 3f_2 - f_1 = 4f_2 \not\geq 0.$$

We also compute that

$$\langle \lambda - \mu, \rho \rangle = \langle f_1 - f_2, \rho \rangle = 1$$

as $f_1 - f_2$ is simple. Putting these together, we obtain

$$d_\lambda(\mu) = q(q^{-1} - 0) = 1.$$

# 4 Proof of proposition 2

## 4.1 A little harmonic analysis

The key theoretical input to prove **proposition 2** is a generalization of the Plancherel theorem from harmonic analysis. Recall the following from the theory of Fourier analysis on locally compact abelian groups:

Let $A$ be a locally compact abelian group and let $A^* = \mathrm{Hom}_{\mathrm{cts}}(A, \mathbf{R}/\mathbf{Z})$ be its Pontryagin dual. Fix a Haar measure $m$ on $A$; then

**Theorem 2: (Plancherel)**

There exists a Haar measure $m^*$ on $A^*$ such that the Fourier transform $L^1(A) \to C_0(A^*)$ restricts to an isometry $L^1(A) \cap L^2(A) \to L^2(A^*)$. Moreover, the Fourier transform extends to an isometric isomorphism $L^2(A) \xrightarrow{\sim} L^2(A^*)$.

We can generalize this to nonabelian groups as follows: let $A$ be a locally compact (not necessarily abelian) group and let $B$ be a compact subgroup; fix a Haar measure $m$ on $A$ (if $B$ is open we may choose the measure such that $m(B) = 1$). Let $\mathcal{H}_{\mathbf{C}}(A, B)$ be the complex Hecke algebra i.e. the $\mathbf{C}$-algebra of continuous compactly supported functions $A \to \mathbf{C}$ which are $B$-bi-invariant, with multiplication given by convolution. We define the set of **zonal spherical functions** (z.s.f.s) on $A$ relative to $B$ as those nonzero $B$-bi-invariant continuous functions $\omega\colon A \to \mathbf{C}$ satisfying

$$\omega(a_1)\omega(a_2) = \int_B \omega(a_1 b a_2)\, dn(b)$$

for all $a_1, a_2 \in A$, where $n$ is the Haar measure on $B$ such that $n(B) = 1$. We say a z.s.f. $\omega$ is **positive definite** if

$$\sum_{i,j=1}^{n} z_i \overline{z_j} \omega(a_i^{-1} a_j) \in \mathbf{R}_{\geq 0}$$

for all $a_1, \dots, a_n \in A$ and $z_1, \dots, z_n \in \mathbf{C}$. It is easily seen that such a function satisfies

- $\omega(1) > 0$;
- $|\omega(a)| \leq \omega(1)$ for all $a \in A$;
- $\omega(a^{-1}) = \overline{\omega(a)}$.

We denote by $\Omega^+$ the set of positive definite z.s.f.s on $A$ relative to $B$; one sees immediately that when $A$ is abelian and $B$ is trivial these are exactly the characters $A \to S^1$. We define for $f \in L^1(A, B)$ ($f \in L^1(A)$ bi-invariant under $B$) the **Fourier transform** $\hat{f} \colon \Omega^+ \to \mathbf{C}$ by

$$\hat{f}(\omega) := \int_A f(a)\omega(a^{-1})\, dm(a);$$

we endow $\Omega^+$ with the weakest topology such that all Fourier transforms $\hat{f}$ of $f \in L^1(A, B)$ are continuous (this coincides with the compact-open topology). It can be shown that this Fourier transform satisfies many of the same properties that the conventional Fourier transform on a locally compact abelian group satisfies. Among these is the following generalization of **theorem 2**:

**Theorem 3: (Plancherel–Godement) [MA26, Thm. 1.5.1]**

There exists a unique positive measure $p$ on $\Omega^+$ (called the Plancherel measure) such that

1. If $f \in \mathscr{H}_{\mathbf{C}}(A, B)$ then $\hat{f} \in L^2(\Omega^+)$;
2. $f \mapsto \hat{f}$ is isometric for the $L^2$ norm on $\mathscr{H}_{\mathbf{C}}(A, B) \subset L^2(A, B)$.

Moreover, the Fourier transform extends to an isometric isomorphism $L^2(A, B) \xrightarrow{\sim} L^2(\Omega^+)$.

When $A$ is abelian and $B = 0$, we note that **theorem 3** reduces precisely to **theorem 2**.

## 4.2 The Proof

For the proof of **proposition 2**, we can make use of the case where $A = T$, $B = T \cap K$.[4] Note that the Hecke algebra $\mathscr{H}_{\mathbf{C}}(A, B)$ in this case is isomorphic to $R(\hat{G}) \otimes_{\mathbf{Z}} \mathbf{C}$ because of the Cartan decomposition[5]. We have, due to Macdonald, an explicit description of the Plancherel measure for this pair. Let $\hat{\Lambda}$ denote the Pontryagin dual of $X^*(\hat{T})$ and take $n$ to be the Haar measure on $\hat{\Lambda}$, normalized so that $n(\hat{\Lambda}) = 1$.

**Theorem 4: (Macdonald) [MA26, Thm. 5.1.2]**

There is an embedding $\hat{\Lambda} \hookrightarrow \Omega^+$ such that the Plancherel measure is concentrated on $\hat{\Lambda}$. Moreover, it is given by

$$p(S) = \int_{S \cap \hat{\Lambda}} \frac{W(q^{-1})}{|W|} \prod_{\alpha \in \Phi^{\vee +}} \left| \frac{1 - s(\alpha)}{1 - q^{-1}s(\alpha)} \right|^2 dn(s) = \int_{S \cap \hat{\Lambda}} \frac{W(q^{-1})}{|W|} \prod_{\alpha \in \Phi^{\vee}} \frac{1 - s(\alpha)}{1 - q^{-1}s(\alpha)}\, dn(s)$$

for $S \subset \Omega^+$ Borel.

Here $W(q^{-1}) := W_0(q^{-1})$ where

$$W_\lambda(q^{-1}) := \sum_{\sigma \in \mathrm{Stab}_W \lambda} q^{-\ell(\sigma)}$$

[4]Note that even though $A$ is abelian, $B$ is nontrivial so we are not in the classical setting of Fourier analysis on A!

[5]and moreover, it was this description of $R(\hat{G})$ as $\mathscr{H}(T, T \cap K)$ that was used to construct the Satake transform in [Gro98, §3]

for $\lambda \in P^+$, $\ell(\sigma)$ the length of $\sigma$ as an element of the Coxeter group $W$. The reader may safely ignore the precise definition of $W(q^{-1})$ and $W_\lambda(q^{-1})$ and treat them as dummy constants; they will eventually cancel out when we compute the formula for $d_\lambda(\mu)$.

*Proof of proposition 2.* Using Macdonald's description of the Plancherel measure, we find via the Plancherel–Godement theorem that

$$\left\langle \mathcal{S}c_\lambda, \mathcal{S}c_\mu \right\rangle = \begin{cases} q^{2\langle \lambda,\rho\rangle} \frac{W(q^{-1})}{W_\lambda(q^{-1})} & \text{if } \lambda = \mu; \\ 0 & \text{else.} \end{cases}$$

This implies by equation (2) that

$$\left\langle \varphi_\lambda, \mathcal{S}c_\mu \right\rangle = d_\lambda(\mu) q^{2\mu-\lambda,\rho} \frac{W(q^{-1})}{W_\mu(q^{-1})} \tag{3}$$

for $\lambda \geq \mu$.

On the other hand, we may use Macdonald's theorem to write

$$\left\langle \varphi_\lambda, \mathcal{S}c_\mu \right\rangle = \frac{W(q^{-1})}{|W|} \int_{\hat{\Lambda}} s(\varphi_\lambda)\overline{s(\mathcal{S}c_\mu)} \prod_{\alpha\in\Phi^\vee} \frac{1-s(\alpha)}{1-q^{-1}s(\alpha)}\, dn(s).$$

One can use

**Theorem 5: Weyl's Character Formula**

For $\lambda \in P^+$,

$$\chi_\lambda = \frac{\sum_{\sigma\in W} \varepsilon(\sigma)(\sigma(\lambda+\rho^\vee) - \rho^\vee)}{\prod_{\alpha\in\Phi^{\vee+}}(1-[-\alpha])}$$

to show that

$$\left\langle \varphi_\lambda, \mathcal{S}c_\mu \right\rangle = q^{\langle \mu,\rho\rangle} \frac{W(q^{-1})}{W_\mu(q^{-1})} \int_{\hat{\Lambda}} \sum_{\sigma\in W} \varepsilon(\sigma) s(\sigma(\lambda+\rho^\vee) - (\mu+\rho^\vee)) \prod_{\alpha\in\Phi^{\vee+}} (1-q^{-1}s(-\alpha))^{-1}\, dn(s).$$

Recall that $\hat{P}(\mu)$ is the generating function in $q^{-1}$ of expressions of $\mu$ as sums of positive roots; hence we have

$$\prod_{\alpha\in\Phi^{\vee+}} (1-q^{-1}s(-\alpha))^{-1} = \sum_{\kappa\in X^*(\hat{T})} \hat{P}(\kappa)s(-\kappa).$$

We now apply this along with the standard property of locally compact abelian groups that

$$\int_{\hat{\Lambda}} s(\lambda-\mu)\, dn(s) = \begin{cases} 1 & \text{if } \lambda = \mu; \\ 0 & \text{else} \end{cases}$$

to obtain

$$\left\langle \varphi_\lambda, \mathcal{S}c_\mu \right\rangle = q^{\langle \mu,\rho\rangle} \frac{W(q^{-1})}{W_\mu(q^{-1})} \sum_{\sigma\in W} \varepsilon(\sigma)\hat{P}(\sigma(\lambda+\rho^\vee) - (\mu+\rho^\vee)).$$

Comparing with equation (3), we find that

$$d_\lambda(\mu) = q^{\langle \lambda-\mu,\rho\rangle} \sum_{\sigma\in W} \varepsilon(\sigma)\hat{P}(\sigma(\lambda+\rho^\vee) - (\mu+\rho^\vee))$$

as claimed. ■

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